

Photocatalytic solar hydrogen production from water on a 100-m² scale

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The unprecedented impact of human activity on Earth's climate and the ongoing increase in global energy demand have made the development of carbon-neutral energy sources ever more important. Hydrogen is an attractive and versatile energy carrier (and important and widely used chemical) obtainable from water through photocatalysis using sunlight, and through electrolysis driven by solar or wind energy^{1,2}. The most efficient solar hydrogen production schemes, which couple solar cells to electrolysis systems, reach solar-to-hydrogen (STH) energy conversion efficiencies of 30% at a laboratory scale³. Photocatalytic water splitting reaches notably lower conversion efficiencies of only around 1%, but the system design is much simpler and cheaper and more amenable to scale-up^{1,2}—provided the moist, stoichiometric hydrogen and oxygen product mixture can be handled safely in a field environment and the hydrogen recovered. Extending our earlier demonstration of a 1-m² panel reactor system based on a modified, aluminium-doped strontium titanate particulate photocatalyst⁴, we here report safe operation of a 100-m² array of panel reactors over several months with autonomous recovery of hydrogen from the moist gas product mixture using a commercial polyimide membrane⁵. The system, optimized for safety and durability, and remaining undamaged on intentional ignition of recovered hydrogen, reaches a maximum STH of 0.76%. While the hydrogen production is inefficient and energy negative overall, our findings demonstrate that safe, large-scale photocatalytic water splitting, and gas collection and separation are possible. To make the technology economically viable and practically useful, essential next steps are reactor and process optimization to substantially reduce costs and improve STH efficiency, photocatalyst stability and gas separation efficiency.

Solar energy is expected to grow in importance as a sustainable energy resource, through local power generation and the operation of commercial solar power plants on a scale of several square kilometres that are currently located at low latitudes where they exploit the available intense solar radiation^{6,7}. Beyond energy generation, the conversion of solar energy into chemicals is also attracting increasing interest because solar energy is often harvested at sites distant from areas with high energy demand and because the intensity of solar radiation fluctuates with time. Photovoltaic-assisted electrochemical, photoelectrochemical and photocatalytic water splitting systems can produce solar hydrogen from water². At present, the most promising of these technologies are photovoltaic-assisted electrolyzers that have reached efficiencies as high as 30% (ref. ³) and been combined with fuel cells to enable end users to generate both solar power and solar fuels to cover

their energy needs throughout the year on a regional scale⁶. While particulate photocatalyst systems exhibit much lower STH values on the order of 1% at most⁸, improvements in photocatalyst performance^{9–12} might make such systems viable and allow their simplicity and scalability to be utilized, provided product gases can be handled safely and efficiently.

Overview of arrayed panel reactor

We explore further scaling and gas handling of solar hydrogen production through photocatalytic water splitting with panel reactors that use photocatalyst sheets^{3,13}. As shown in Fig. 1 and Extended Data Fig. 1, a 100-m² scale prototype photocatalytic solar hydrogen production system was built at the Kakioka Research Facility within the University

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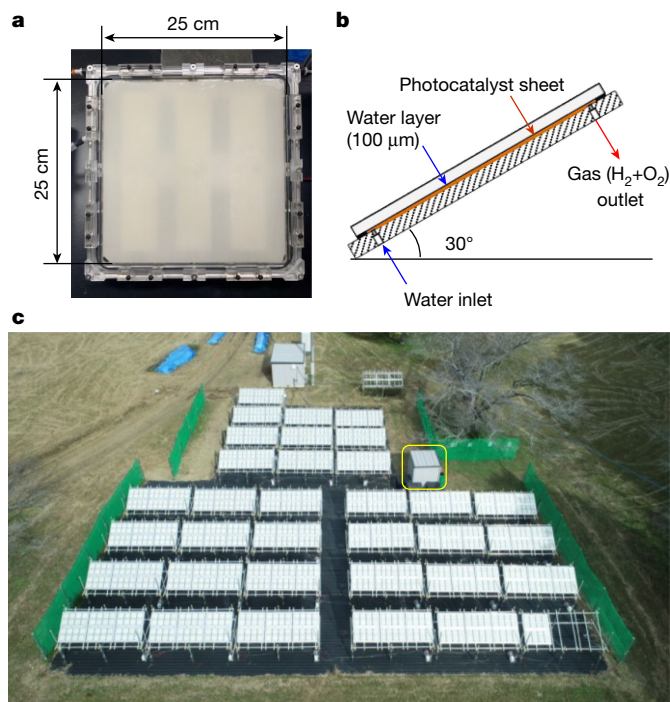


Fig. 1 | The 100-m² water splitting photocatalyst panel reactor.

a, A photographic image of a panel reactor unit (625 cm²). **b**, The structure of the panel reactor unit viewed from the side. **c**, An overhead view of the 100-m² solar hydrogen production system consisting of 1,600 panel reactor units and a hut housing a gas separation facility (indicated by the yellow box).

of Tokyo by arraying 1,600 reactor units. Each unit has a light receiving area of 625 cm², with the gap between the ultraviolet-transparent glass window and photocatalyst sheet adjusted to 0.1 mm (see Fig. 1a, b and Extended Data Fig. 1a) to minimize the water load and prevent accumulation and ignition of the product oxyhydrogen gas. For a diagram and photographic images of a 3-m² module, consisting of 48 panel reactor units, see Extended Data Fig. 1b–d. In this system, gas product transport and reactant water transport use polyurethane tubes with inner diameters of 8.6 mm and 4.0 mm, respectively. An overhead view of the 100-m² water splitting photocatalyst panel reactor array, consisting of 33 $\frac{1}{3}$ modules of 3 m² each, is provided in Fig. 1c.

Photocatalyst sheet

Photocatalyst sheets were prepared on clear glass sheets manually, and on frosted glass sheets using a programmed spray system (see Methods). The former sheets were installed in August 2019 and used until July 2020, and replaced with the latter sheets afterward without changing the other system parts. After application, the particulate layer covered the entire surface of the glass sheet to a thickness that varied from 4 to 10 µm across the sheet (see Fig. 2 and Extended Data Fig. 2a, b). Upon increasing the loading of the photocatalyst layer, the water splitting rate quickly saturated (Extended Data Fig. 3). The photocatalyst layer contained modified SrTiO₃:Al particles several hundred nanometres in size, and immobilized by silica nanoparticles that formed mesoporous channels in the interparticle voids (Fig. 2c and Extended Data Fig. 2c).

Before constructing the large-scale panel reactor array, we conducted an indoor acceleration test using a small panel reactor under continuous exposure to simulated standard sunlight (AM 1.5 G, 1 kW m⁻²)⁴. After an activation period of several days, small photocatalyst sheets (5 cm × 5 cm) fabricated on clear, flat glass were found to split distilled water into hydrogen and oxygen with an STH efficiency of 0.48% (Extended Data Fig. 4a). The STH efficiency decreased gradually over

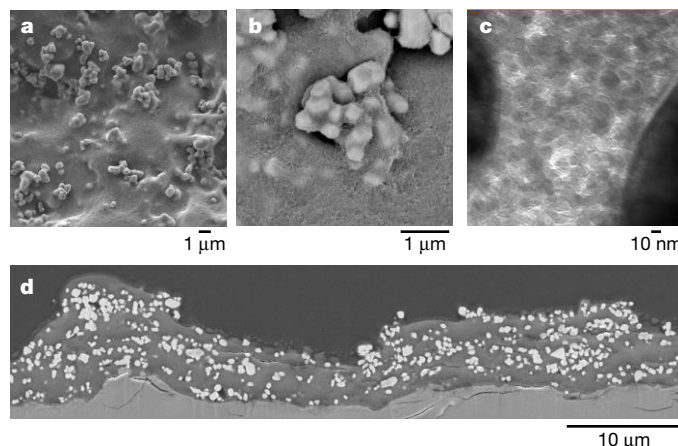


Fig. 2 | Electron microscopy images of photocatalyst sheets. **a**, **b**, Top-view secondary electron (**a**) and backscattered electron (**b**) images acquired by scanning electron microscopy. **c**, A cross-sectional image acquired by transmission electron microscopy. **d**, A cross-sectional backscattered electron image acquired by scanning electron microscopy. These specimens were fabricated by depositing the photocatalyst on frosted glass sheets using a program-controlled sprayer.

time and dropped below 0.40% in 280 h. Photocatalyst sheets fabricated on the frosted glass were more active and durable (Extended Data Fig. 4b), reaching after activation an STH efficiency of 0.51% that remained above 0.40% over 1,600 h (a timespan equivalent to seven months under an outdoor insolation of 7.6 kWh m⁻² per day, or to almost one year under the climate conditions at the Kakioka Research Facility). The activation period exhibited by the system probably results from relocation of impregnated chromium components in the form of Cr(III) species onto Rh sites on the surface of the SrTiO₃:Al (ref. ¹⁴). Deactivation probably results from deterioration of this co-catalyst during long-term operation¹², given that the microstructure of the photocatalyst layer remained intact even after field testing over approximately six months (see Extended Data Fig. 2d).

Bubbles of oxyhydrogen gas 0.1–0.6 mm in size formed on the photocatalyst sheet surface (Supplementary Video 1), similar to the nucleation of bubbles on hydrophobic pits during water electrolysis¹⁵. It is probable that hydrogen and oxygen migrated through the mesoporous channels between the hydrophilic silica nanoparticles in a supersaturated liquid phase and then formed bubbles on the surface of the photocatalyst sheet. With a gap between photocatalyst sheet and reactor as wide as 1 mm, many small bubbles formed⁴. But when the photocatalyst sheet was tightly contained in the reactor unit and illuminated, oxyhydrogen gas bubbles coalesced and minimized both the liquid/gas interface and associated scattering of incident light (Supplementary Video 2). Despite the narrow headspace of 0.1 mm, the reactor unit design smoothly expelled gas bubbles.

Arrayed panel reactor

The trial photocatalyst panel reactor, which comprised three modules with a total light receiving area of 9 m², generated moist oxyhydrogen gas at a rate of 568 ml min⁻¹ (at 25 °C and 100 kPa, excluding the saturated water vapour volume) when exposed from 11:00 to 11:30 am on 29 August 2020 (the end of summer) to natural sunlight with an irradiance intensity of 0.88 kW m⁻². The gas generation rate was equivalent to an STH value of 0.76%.

The 100-m² photocatalyst panel reactor array, using photocatalyst sheets prepared on frosted glass, was then operated from 22 September to 21 December 2020 (from autumn to winter). Extended Data Figure 5 plots continuous records of sunlight intensity, ultraviolet power,

amount of humid oxyhydrogen gas produced, daily STH efficiency and the ultraviolet proportion of the solar irradiation over this period. On 22 September 2020, the peak production rate of $3.6\text{--}3.7\text{ l min}^{-1}$ occurred from 11:00 to 11:30 am at an outdoor temperature of $34\text{ }^{\circ}\text{C}$. Supplementary Video 3 illustrates collection of the product gases generated in the 100-m^2 photocatalyst panel reactor array under natural sunlight irradiation on 30 September 2020, before the hydrogen separation membrane was installed. The STH value reached with the reactor system under natural sunlight decreased gradually over time, to approximately 0.3% by the middle of December 2020. Given the durability of the photocatalyst sheet (see Extended Data Fig. 4b), we attribute the fall in performance to changes in the weather conditions that reduced the proportion of ultraviolet radiation in the total irradiance in natural sunlight at the Kakioka Research Facility from approximately 5% to 2% (Extended Data Fig. 5f). Note that the $\text{SrTiO}_3\text{:Al}$ photocatalyst employed is active only under ultraviolet light.

In an earlier field test, the 100-m^2 panel reactor using photocatalyst sheets on flat, clear glass continuously produced oxyhydrogen gas over eight months, between November 2019 and July 2020. The same panel reactor, after replacing initial sheets with photocatalyst sheets prepared on frosted glass, was operated from July 2020. In both configurations, the daily STH efficiency dropped after December. We attribute this to degradation and detachment of the photocatalyst particle layer caused by repeated freezing and melting of the water in the system. From late December to February, the water in the panel reactors, tubes and gas reservoir was observed to repeatedly freeze overnight and then to melt in the daytime. Once the photocatalyst particle layer was peeled off the sheet, the photocatalyst settled at the lower part of the reactor unit and did not effectively receive sunlight.

Gas separation unit

Extended Data Figure 6 shows the gas-separating membrane unit into which the moist, gaseous oxyhydrogen product from the 100-m^2 photocatalyst panel reactor was fed via gas collection and transportation tubes (Fig. 1b). Two alternately filled gas reservoirs (each 3 l) were situated just prior to the membrane separator. The filter element comprised a bundle of hollow polyimide fibres assembled in a stainless steel cartridge⁵. The permeability of these fibres with regard to hydrogen is lower than that of water vapour but about ten times higher than that of oxygen. The membrane separator worked based on suction provided by a diaphragm pump such that the pressure difference across the membrane was almost 100 kPa. The capacity of the membrane unit and the diaphragm pump to process the gaseous product exceeded the gas evolution rate, such that the unit only had to function intermittently to separate hydrogen (Fig. 3a). The hydrogen-enriched filtrate gas was expelled by the diaphragm pump at atmospheric pressure while the residual oxygen-enriched gas was expelled from the filter cartridge.

Figure 3b plots the accumulated amounts of feed, filtrate and residue gases over the course of the entire clear day on 2 October 2020, where the results of occasional gas composition analyses are provided along the filtrate and residue curves. The variations of solar illumination intensity and the gas evolution rate in the water splitting panel reactor are also shown in Fig. 3c. The gas production rate from the photocatalyst panel reactors follows the fluctuation of the weather conditions quickly. The total feed of humid oxyhydrogen gas on the day was 970 l, collected at an atmospheric pressure of 106.9 kPa and ambient temperature of $24.4\text{ }^{\circ}\text{C}$ as averaged over the daytime. Because the feed gas contains hydrogen and oxygen at the molar ratio of 2.0 and water at its saturation vapour pressure (approximately 3.0 kPa), the collected gas contains a total 27.2 mol hydrogen. The membrane separation gave 505 l of hydrogen-enriched filtrate gas saturated with water vapour, with a hydrogen purity exceeding 94% (excluding water vapour) on average and thus a hydrogen yield of 19.9 mol. Accordingly, hydrogen recovery reached 73% even though the commercial membrane we used

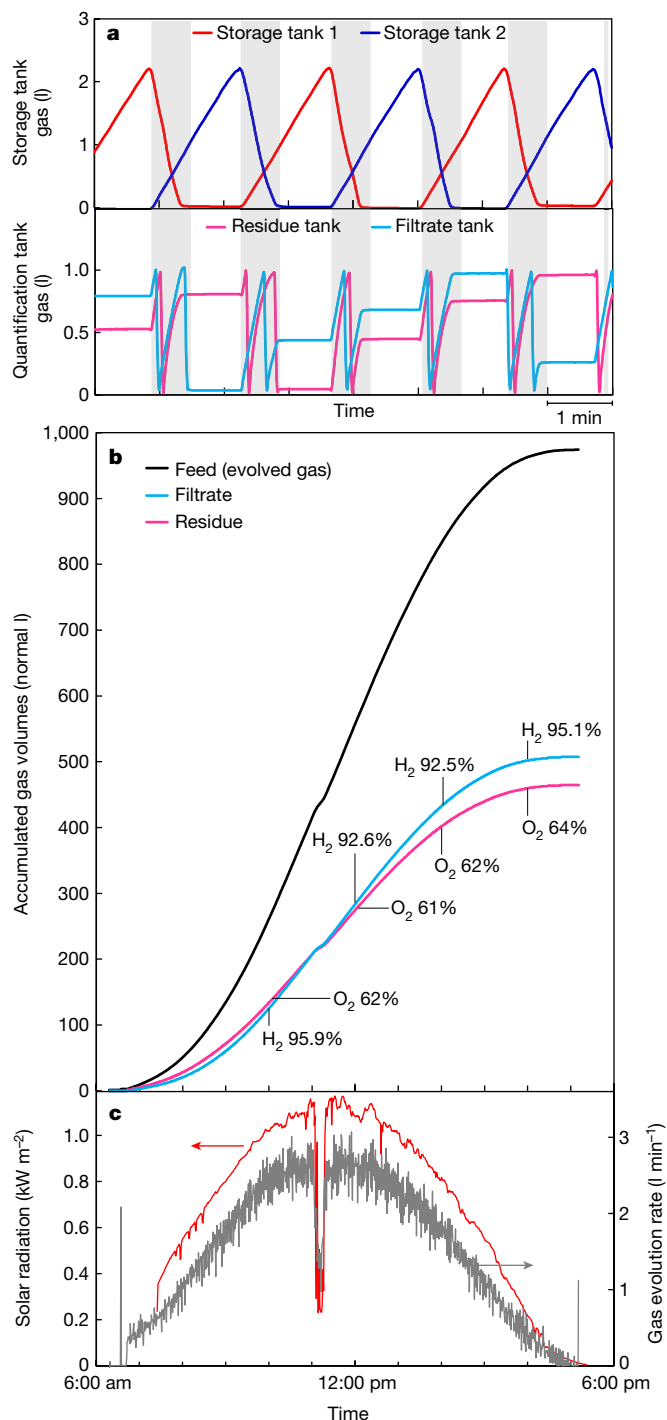


Fig. 3 | Performance of the gas separation unit connected to the 100-m^2 water splitting photocatalyst panel reactor on 2 October 2020 at the Kakioka Research Facility. **a, Time courses of the gas volumes contained in the storage and quantification tanks. The periods during which the filter cartridge was operational are shaded. **b**, The accumulated amounts of moist oxyhydrogen feed, filtrate and residue gases. The ratios of hydrogen and oxygen in the filtrate and residue gases excluding water vapour are notated along the curves, respectively. **c**, The variations of solar illumination intensity (red) and the gas evolution rate in the water splitting panel reactor (grey).**

is not designed for hydrogen separation from oxygen. Throughout field testing, the gas-separating membrane unit functioned without signs of deterioration and regardless of the varying production rate of humid oxyhydrogen gas at different weather conditions (Extended Data Fig. 7).

Safety issue

The entire hydrogen production system operated under field conditions for more than a year without spontaneous explosion or any other failure occurring. For a more stringent safety test, we subjected each component of the solar hydrogen production system to intentional ignition of the oxyhydrogen gas at the hydrogen-to-oxygen ratio of two and saturated with water vapour (see Methods). A large section of the photocatalytic water splitting reactor array, with a 70-m² light receiving area and operating under sunlight, remained undamaged when the gaseous product in the connected gas collection tube was intentionally ignited (Extended Data Fig. 8 and Supplementary Video 4). The vinyl tube with an inner diameter of up to 20 mm also remained undamaged when moist oxyhydrogen gas inside the tube was intentionally ignited and subsequently detonated. However, tubes with larger inner diameters would burst. The detonation inside a 10-mm inner diameter tube is shown in Supplementary Videos 5 and 6. The hollow polyimide fibre membrane separator was also undamaged and it retained its gas separation performance following detonation of the oxyhydrogen gas introduced into the gas separation unit. However, intentional ignition and the resultant deflagration/detonation damaged the gas reservoir tank submerged in water, in most instances rendering the tank unusable. Installation of a spiral partitioning plate inside the tank reduced the impact of the explosion and ensured the tank remained usable without deteriorating of its function. Taken together, the intentional ignition results suggest that oxyhydrogen gas can be handled safely provided the gas is confined in a narrow passage in each compartment and transported using appropriate tubing.

Challenge for practical use

Our findings demonstrate that scaling of solar hydrogen production via photocatalytic overall water splitting to a size of 100 m²—by far the largest solar hydrogen production unit yet reported to our knowledge—is feasible, with further scaling in principle possible without efficiency degradation. But even though our system has delivered the highest solar hydrogen output to date due to its large size¹⁶, efficiency is low and the STH value well below what can be achieved with photovoltaic-assisted water electrolyzers³.

For photocatalytic water splitting to be practically relevant, the need for better photocatalysts that utilize visible light remains a fundamental issue. The SrTiO₃:Al photocatalyst used in this study works in the ultraviolet range and under natural sunlight reaches an STH of only 0.76%. Although another photocatalyst is available that reaches at ambient pressure a higher STH of 1% (ref. ¹⁷), STH would need to reach 5–10% to realize economically viable solar hydrogen production¹. And although present photocatalyst sheets were stable over several months during bench testing and degraded photocatalyst sheets can be readily exchanged during operation, but a longer lifetime of a few years is desired.

The present panel reactor was built with a focus on sufficient robustness to ensure long-term outdoor operation and without considering production and operating costs, but any practically relevant solar fuel production system must be cost competitive with existing fuel production technologies^{1,18}. Simpler reactors made of lightweight and inexpensive materials that still ensure safety and durability will need to be developed. The finding that the present system was unable to produce hydrogen with a net energy gain (see Methods) highlights that the power requirements for water and gas processing will need to be

reduced. The current hydrogen recovery system based on a commercially available membrane gives hydrogen that is still in the explosive range (4% < H₂ < 95%)¹⁹ and leaves more than 20% of the hydrogen in the feed gas. In addition to optimization of the gas processing and operating conditions, using specially designed pumps, the development of membranes with better hydrogen permeability and lower oxygen permeability is clearly essential.

In closing, we note that although there were no indications of critical issues and hazards during our preliminary safety tests, there are obvious safety issues associated with operating of the entire solar hydrogen reaction panel system and any newly developed units should be inspected and approved by chemical engineering safety experts to comply with safety and legal liability requirements.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-021-03907-3>.

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Methods

Preparation of photocatalyst sheets

The SrTiO₃:Al photocatalyst was mass-produced in quantities up to 2 kg by the SrCl₂ flux method⁴. Cocatalysts were deposited by impregnation in aqueous solutions of Na₃RhCl₆, Cr(NO₃)₃ and Co(NO₃)₂ followed by calcination in air. The loading amounts of Rh, Cr and Co were 0.1 wt% each with respect to the SrTiO₃:Al photocatalyst. SrTiO₃:Al (50 g) was dispersed in aqueous solution (approximately 120 ml) with stirring and ultrasonication for five minutes. Then, the water solvent was evaporated with a rotary evaporator set at 80 °C in approximately 30 min. The resulting solid was dried at 60 °C for 40 min and mildly crushed for a few minutes. Then it was divided into ten porcelain plates evenly and calcined at 350 °C for 1 h with a heating rate of 10 °C min⁻¹. This method allowed for mass production of modified SrTiO₃:Al with high durability although the apparent quantum yield of this photocatalyst processed into sheets in overall water splitting was approximately 50% at 365 nm, inferior to that of the suspension of the state-of-the-art SrTiO₃:Al photocatalyst site-selectively loaded with cocatalysts by photodeposition⁹.

The photocatalyst sheets used in the earlier version of the panel reactor (installed from August 2019 and operated until July 2020) were prepared by spraying a slurry of modified SrTiO₃:Al mixed with chained colloidal silica (ST-0UP, Nissan Chemical Corp.) and calcium chloride onto flat, clear glass sheets (G-leaf, 250 mm × 250 mm × 0.1 mm thick, Nippon Electric Glass) in ambient air. In this process, the modified SrTiO₃:Al, chained colloidal silica and calcium chloride were dispersed in 40 ml of pure water at a 4.0:1.0:0.046 mass ratio. The resulting suspension was sprayed over the entire glass sheet (which was maintained at a temperature of 70 °C) using a manual sprayer (HP-BCN, Anest Iwata), applying ten passes. In the later version of the panel reactor (starting operation from July 2020), sand-blasted frosted glass sheets (OA-11, 250 mm × 250 mm × 0.4 mm thick, Nippon Electric Glass) were employed in expectation of enhancing the wettability to the sprayed suspension and the mechanical strength of the photocatalyst particle layer. Spray valves (SV-91, SAN-EI TECH, Ltd) operated by a custom-made programmable X-Y robot were used to prepare the photocatalyst sheets. The modified SrTiO₃:Al, chained colloidal silica and calcium chloride (at a 4.0:1.0:0.046 mass ratio) were dispersed in 200 ml of pure water. The suspension was sprayed onto the frosted glass sheets (maintained at 70 °C) using the program-controlled sprayers, applying 20 passes. The photocatalyst sheets were finally heated at 90 °C at a relative humidity of 50% for 4 h. In both cases, the modified SrTiO₃:Al loading was held constant at 0.89 mg cm⁻². The rate of photocatalytic water splitting as measured under irradiation from an ultraviolet light emitting diode array (NCSU276AT, λ = 365 nm, Nichia Corp.) was found to plateau at higher loadings (Extended Data Fig. 3).

The evolution of gas bubbles on a photocatalyst sheet was observed using a digital microscope (VHX-5000, KEYENCE) equipped with a zoom lens (VH-Z50L, KEYENCE) with the sheet immersed in pure water and tilted at an angle of 45°. During this experiment, the sheet was irradiated vertically with a Xe lamp. The zoom lens of the microscope was placed horizontally such that the evolution of bubbles was observed from an angle of 45° relative to the photocatalyst sheet surface.

Fabrication of water splitting panel reactors

Specific design criteria have been established based on lab-scale experiments involving the ignition of oxyhydrogen gas at atmospheric pressure¹³. To further assess such design criteria in a field environment in a long term, reactor units with the light accepting area of 625 cm² (25 cm × 25 cm) were fabricated, and the 100-m² reactor array was constructed at the Kakioka Research Facility within the University of Tokyo (situated at a longitude of 140.32°E and latitude of 36.38°N with an elevation of 34 m relative to sea level). The size of the reactor units was chosen considering a balance between the ease of manual construction work in the laboratory and the availability of human resources and time

to install a 100-m² scale reaction facility. In future, all the components should be designed considering the cost effectiveness, safety issues and other factors by the experts.

The photocatalyst sheets were assembled into a reactor unit (Fig. 1 and Extended Data Fig. 1a) with a body made of transparent polycarbonate formed by injection moulding. An ultraviolet-transparent glass window (Glanova, 2.8 mm thick, Nippon Sheet Glass) was sealed onto the reactor body using a rubber packing lubricated with silicon oil. In the later version of the reactors, the device was sealed with unlubricated rubber O rings. The fluid inlet/outlet holes were made with connectors that locked thin plastic tubes (inner diameter of 1.6 mm) into place. The gap between the window and photocatalytic sheet was designed to be 0.1 mm to minimize the dead volume. A frame-and-skin structure was employed to obtain the required stiffness, in conjunction with reinforcement by aluminium alloy pillar backing frames to maintain the desired 0.1-mm gap. The reaction system was operated in an ambient atmosphere, and plastic elements with a pressure tolerance of 1 MPa were employed for piping and fitting. Nevertheless, high internal pressure values were found to result in gas and water leakage at the rubber seals, and the maximum allowable pressure difference with respect to the atmospheric pressure was determined to be 5 kPa.

In this work, 48 reactor units were assembled into a module to provide a light receiving area of 3 m², supported by a single scaffold. The scaffold was built with aluminium alloy tubes fixed on steel piles driven into the ground. The racks for the reactors were directed toward the south and tilted upwards at an angle of 30°.

A total of 33¹/₃ such modules (equivalent to 1,600 reactor units) were assembled, and the arrangement of these modules in the experimental field trials is shown in Fig. 1c. The gaseous oxyhydrogen product was transferred from each reactor unit through a Teflon tube with an inner diameter of 1.6 mm and subsequently passed through polyurethane tubes with an inner diameter of 8.6 mm, using plastic connectors daisy-chained across the whole panel reactor. These polyurethane tubes also served to drain off overflow water into the water tank. These tubes were connected to a gas booster pump to stabilize the flow rate of the gaseous products and finally to a gas flowmeter or filtration apparatus.

A reactant water feeding system was installed as a single pump assembly for every module, supplied with ion-exchanged water originally taken the local municipal supply. This system was composed of a water tank, a small diaphragm feed pump, automatic valves and water level sensors and, was capable of fulfilling the water splitting panel reactor units with water within 10 s once a day. Each module had four levels of reactor units, and four automatic valves were connected to the bottom water inlets to prevent the backflow of gas. The gaseous oxyhydrogen produced by the module was separated from overflow water in the polyurethane tubes, with the water being drained into the water tank.

Fabrication and operation of gas-separating membrane units

The humid oxyhydrogen gas from the photocatalyst panel reactor was temporarily stored and then intermittently released to a gas-separating membrane unit to reduce the equipment required and the energy consumption of the system. The membrane unit for hydrogen recovery consisted of a hydrogen-permeable filter element made of commercial polyimide hollow fibres⁵, a vacuum pump for filtrate suction, a dual gas storage tank unit for gas feed rate control and quantification tanks for the filtrate and residual gases (Extended Data Fig. 6). The two quasi-isobaric gas storage tanks comprised cylindrical vessels (3 l) fixed in a tub of water and were operated alternately and seamlessly by a dedicated control circuit. The circuit monitors the gas volumes in the tanks by electronic linear level gauges dedicated for each and controls the automated valves for filling and releasing of the gaseous oxyhydrogen product. The two storage tanks exhibit alternative sawtooth filling level curves, and the downslopes of these curves correspond to the governed gas feed speed to the filter (Fig. 3a). The feed rate from the storage tanks to the filter element was regulated with a throttle valve.

The feed gas was provided to one end of the outer shell containing the hollow fibres in the filter cartridge through a single polyurethane tube with an inner diameter of 8.6 mm and the residual oxygen-enriched gas was emitted from the other end (Fig. 3a). The interior of the device was connected to a vacuum pump that removed the hydrogen-enriched filtrate gas at the end near the feed gas inlet and the other end of the unit was closed. The effluent from the vacuum pump was therefore the hydrogen-enriched product at atmospheric pressure. The filter cartridge was sized such that it was able to process the maximum anticipated gas production rate of the present photocatalyst panel reactor (6 l min^{-1}) while providing the highest possible purity. The volumes of the gas exiting from the unit (the residual gas) and the vacuum pump exhaust (the treated gas) were measured based on the level heights in the two dedicated quantification tanks. The residing volumes of the residue tank and filtrate tank independently draw sawtooth curves during the gas-releasing periods on the feed gas storage tanks (Fig. 3a). The compositions of both gas streams were alternately analysed using a quadrupole mass spectrometer calibrated with a 2:1 mixture of hydrogen and oxygen from a water electrolyser. The mass spectra indicated that nitrogen was not detected during the reaction and thus there were no air leaks in the reactor or filtration apparatus. The filter cartridge, vacuum pump and mass spectrometer were housed in a dedicated hut. In contrast, the water tubs containing the reactor gas storage tanks and the gas quantification tanks were placed outdoors in the vicinity of the hut, to allow for the rapid dilution of the effluent oxyhydrogen in the ambient air. To ensure operator safety, the entire filtration apparatus was electronically remote-controlled and monitored at a distance of 30 m from the hut.

The gas-separating membrane units separated hydrogen with purity greater than 90% from the moist gaseous oxyhydrogen product over the seasons as shown in Fig. 3b and Extended Data Fig. 7. However, during the winter season, the hydrogen recovery proportion was lower (Extended Data Fig. 7a). This is probably because the permeability of the polyimide fibre membrane was lowered at lower temperatures, resulting in an increase of the residue gas proportion. By matching the feed gas rate and the permeability using the throttle valve, the hydrogen recovery proportion could be improved without deteriorating the hydrogen purity as shown in Extended Data Fig. 7b.

Performance evaluation of photocatalyst panel reactors

The ability of the panel reactor unit to produce oxyhydrogen gas at a sufficient rate was examined under ultraviolet light ($\lambda = 365 \text{ nm}$) generated by a light emitting diode array (Supplementary Video 2). In these trials, reactant water could be transferred into the unit using a syringe connected to a lower inlet to allow sufficient control of the water level in the reactor. In addition, the durability of the photocatalyst sheets was examined using a small panel reactor design reported previously⁴. The panel reactor was continuously irradiated with simulated sunlight (AM 1.5 G) with a solar simulator (XES-40S1, SAN-EI ELECTRIC).

The outdoor panel reactor photocatalysis tests were performed in a stepwise manner during the construction of the entire equipment considering the weather conditions. Specifically, on 6 September 2019, the product flow from a 3-m^2 light receiving area was monitored using a calibrated mass flowmeter, while on 19 September 2019, the output from a 30-m^2 light receiving area was assessed by bubbling the effluent gas into a water reservoir. During November and December 2019, the outgoing gas flow from the entire 100-m^2 light receiving area was measured frequently during the daytime. In addition, tests of the gas filtration system in the 100-m^2 reactor array were performed from January to May 2020. These trials all employed photocatalyst sheets prepared on flat, clear glass using manual spraying. In July 2020, the photocatalyst sheets were started to be replaced with those prepared on frosted glass using a program-controlled sprayer. The product rate was measured by a soap-film flowmeter with automatic correction by referring to the atmospheric pressure, water vapour pressure, and

temperature (Horiba, model SF-2U/VP-4U). Insolation, ultraviolet intensity and temperature were acquired from a weather station (Vantage Pro II Plus, Davis Instruments), a pyranometer (SR30, Hukseflux), and a total ultraviolet radiometer (CUV5, Kipp & Zonen). The weather conditions and the gas production rate were periodically recorded from 20 September 2020 by an Internet-aided data logging setup. These data were gathered at a sampling rate of 1 min^{-1} and processed by a microcomputer and transmitted via the Internet to our laboratory, and the acquired data are presented in Extended Data Fig. 5. Note that the oxyhydrogen gas was not produced in late December 2020, because of the cold weather freezing the panel reactor units and other equipment. On 2, 6 and 30 October the gas was directed to the gas separation equipment, without delivery to the gas flowmeter. On 9 and 10 October, the daily STH values were abnormally high. These values are not reliable because of bad weather and the low solar radiation. The data on 18 November and 8 December were not reliable, either, owing to malfunction of the soap-film flowmeter.

The STH values were calculated according to the equation:

$$\text{STH} = \frac{\text{Output energy as H}_2}{\text{Energy of incident solar light}} = \frac{r_{\text{H}_2} \times \Delta G_r}{P_{\text{sun}} \times S},$$

where P_{sun} is the energy flux of the incident sunlight, S is the irradiated area, r_{H_2} is the rate of hydrogen production and ΔG_r is the Gibbs energy change for the reaction (237 kJ mol^{-1} for overall water splitting under standard conditions at 298 K). In the calculation of the daily STH values under natural sunlight, the daily amounts of solar radiation and oxyhydrogen gas collected in the gas storage tank unit were considered. The energy flux of the incident sunlight was obtained from the weather station instrumentation and the pyranometer.

The hydrogen recovery proportion was calculated according to the equation:

Hydrogen recovery proportion

$$\begin{aligned} &= \frac{\text{Amount of recovered hydrogen}}{\text{Amount of generated hydrogen}} \\ &= \frac{\text{Amount of filtrate gas} \times \text{Concentration of hydrogen}}{\text{Amount of generated gas} \times \text{Concentration of hydrogen}} \end{aligned}$$

Both the filtrate gas and generated gas were at the atmospheric pressure and contained hydrogen, oxygen and water at the saturated vapour pressure. Therefore, the sum of the partial pressures of hydrogen and oxygen was equal to the pressure obtained by subtracting the water vapour pressure from the atmospheric pressure. The ratio of hydrogen to oxygen in the gaseous product mixture was confirmed to be 2.0 and that in the filtrate gas was measured by mass spectroscopy. Note that the hydrogen-to-oxygen ratio of the product gas was two because no sacrificial reagent was used.

The balance of output and input energy is important in energy applications. Based on the one-day operation on 2 October (Fig. 3), the power supplied to the vacuum pump, which was the major part of energy consumption within the whole system, was estimated. The pump operation period was approximately 38 s to empty 2.4 l of the tank active volume. The produced oxyhydrogen gas (970 l) was then filtered in $404 (= 970/2.4)$ cycles of operation. The pump power was 400 W at maximum. These numbers yield a one-day power consumption of $1.7 \text{ kWh} = 6.1 \text{ MJ}$. On the other hand, the amount of hydrogen energy produced on the same day was 5.0 MJ . Therefore, the present reaction system did not convert solar energy into hydrogen energy in the net, and improvement in the gas separation apparatuses and procedure is essential.

Examination of safety issues

Experiments involving the intentional ignition of the oxyhydrogen gas in each component, such as reactor panels, tubes, gas reservoirs and

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gas separation unit, were performed in the Kakioka Research Facility with permission from Environment, Health, and Safety Office of School of Engineering of the University of Tokyo. In each case, the item being examined was filled with oxyhydrogen gas produced by a water electrolyser or the photocatalytic water splitting panel reactor, and this mixture was ignited by a spark across an electrode distance of 0.5 mm generated by applying a 15-kV alternating current across the electrodes for 1 s. The test specimens included soft polyvinyl chloride tubes (inner diameter 10 or 20 mm, length 0.3–100 m), polyurethane tubes (inner diameter 8 mm, length 1–15 m), the commercial polyimide gas filter, polycarbonate gas tanks (volume 3–4 l) immersed in water and the photocatalytic unit reactor panel (625 cm²). In most cases, the inner volume of the item was purged with an oxyhydrogen flow with the exit from the item immersed in a water tub through which the outgoing gas bubbled. This same test was also performed with a large section of the reactor array (with a 70-m² light receiving area) under daytime operation conditions by igniting the gaseous product at the exit of the tube carrying the collected gas, with the other end of the spark gap tube immersed in water through which the gas bubbled. These experiments were recorded using a video camera at both regular speed (30 frames s⁻¹) and high speed (1,000 frames s⁻¹). When the evolved gas was confined in a layer 0.1 mm thick in the panel reactor unit and were transported using polyurethane tubes with an inner diameter of 8.6 mm, any resulting deflagration and detonation were harmless. In the case of soft polyvinyl chloride tubes with an inner diameter of 10 mm, the propagation speed of the reaction of oxyhydrogen gas was estimated to be $2.8 \times 10^3 \text{ m s}^{-1}$ from Supplementary Video 6, indicative of the occurrence of detonation. However, the tube was undamaged and used repeatedly. Similarly, soft polyvinyl chloride tubes with an inner diameter of 20 mm were undamaged when the inner oxyhydrogen gas was detonated. The gas separation unit made of hollow polyimide fibres was also found to be undamaged, under the introduction of detonation from the inlet of the separation system.

Disclaimer

The experimental procedures and results described herein involved the generation of gaseous oxyhydrogen or the separate generation of hydrogen and oxygen. These gases are either flammable (hydrogen and oxyhydrogen) or promote combustion (oxygen) and so various regulations may apply to such work depending on the administration of the experimental area. The authors did not observe unintended ignition or spontaneous combustion of these gases in trials of the prototype

photocatalytic solar hydrogen production system. However, the results of experiments involving the intentional ignition of oxyhydrogen gas volumes in excess of several hundreds of millilitres at atmospheric pressure demonstrated that the resulting explosion can be potentially hazardous. Therefore, the authors do not recommend performing the experiments described herein without well-designed measures to avoid and control such hazards. Prior to such research, professional safety experts should be consulted to properly prepare the experimental space and to plan any field work involving these gases.

Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

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Author contributions K.D. planned and supervised this project. H.N. and T.Y. were responsible for the basic design, on-site design, construction and supervision of the plant. H.N. and T.T. investigated the cocatalyst deposition process, and designed, tested, fabricated and supervised the mass production of the photocatalytic reactors. S.O. and H.T. prepared the SrTiO₃:Al photocatalyst on a large scale. M.Y., Y.K., Y.N., R.N. and T.H. prepared the modified photocatalyst. M.N. and N.S. performed electron microscopy. Y.M. examined in situ gas evolution. T.Y. designed and constructed the gas filtration apparatus. T.Y. and H.N. performed the field testing of the entire solar hydrogen production system. H.N., T.Y., S.A., T.W. and K.D. examined the safety issues. T.Y., H.N., T.H. and K.D. discussed the results and wrote the manuscript.

Competing interests H.N., T.Y. and K.D. of the University of Tokyo hold patents related to this work (Japanese Unexamined Patent Application publication no. 2021-75,319).

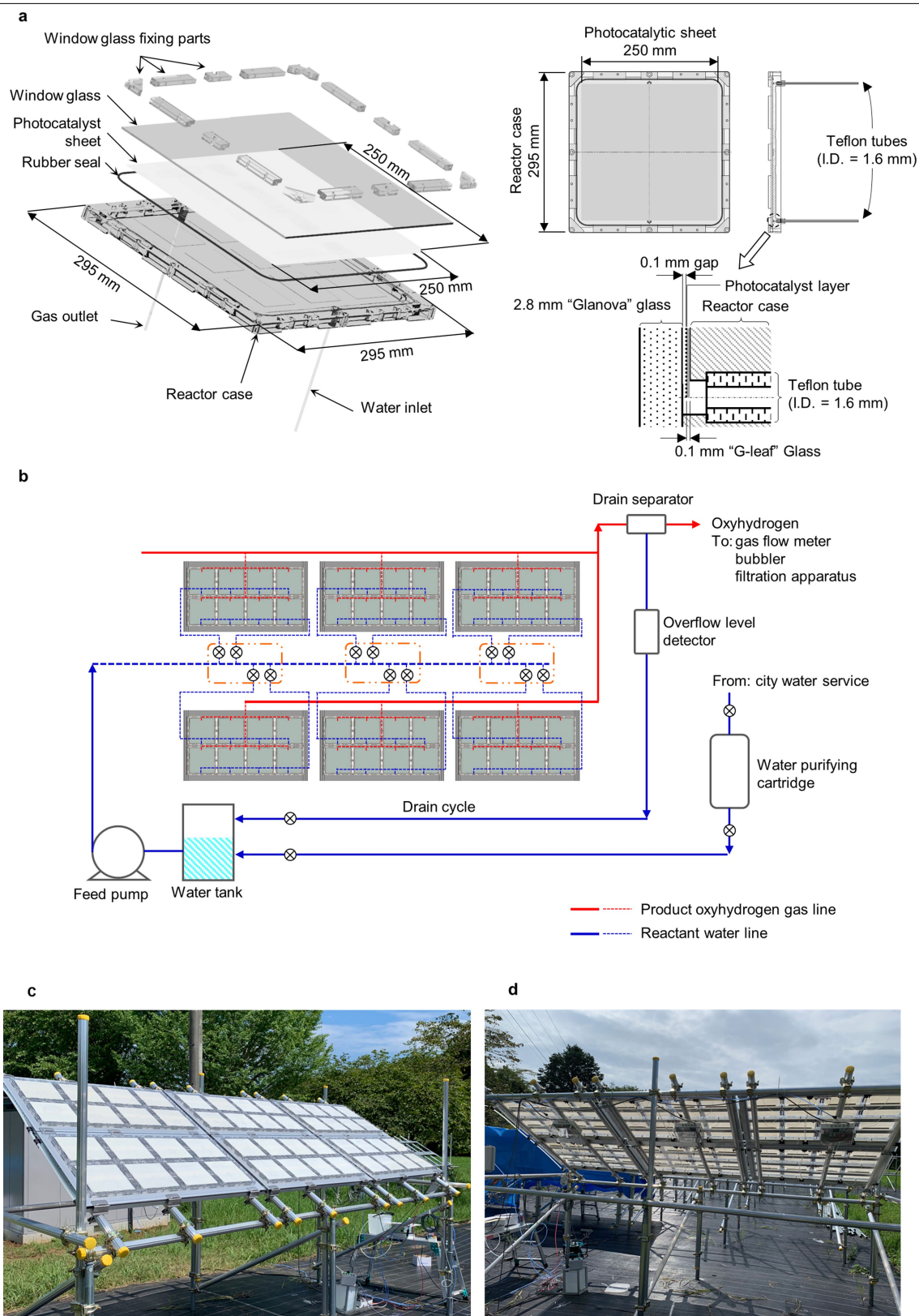
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-021-03907-3>.

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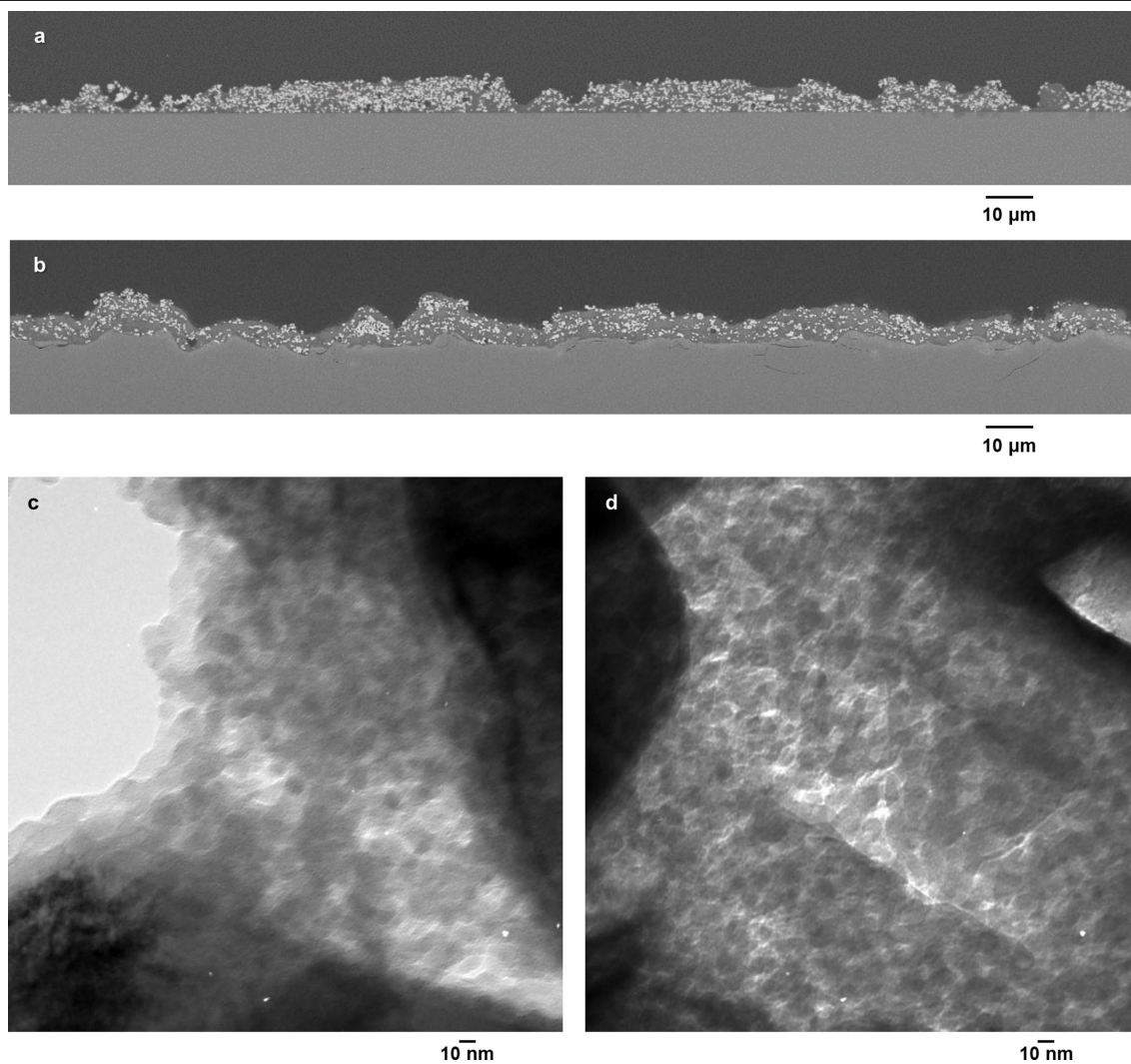
Peer review information Nature thanks Frank Osterloh and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Extended Data Fig. 1 | The photocatalyst panel reactors. **a**, The design of a panel reactor unit (625 cm²). **b**, A diagram showing the structure.

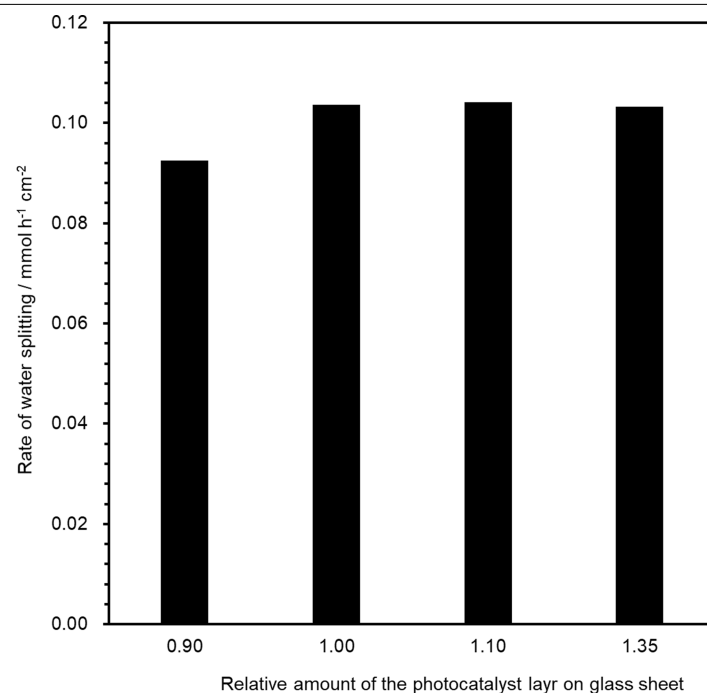
c, d, Photographic images of the 3-m² module consisting of 48 panel reactor units viewed from the front (**c**) and rear (**d**).



Extended Data Fig. 2 | Photocatalyst sheet microstructures.

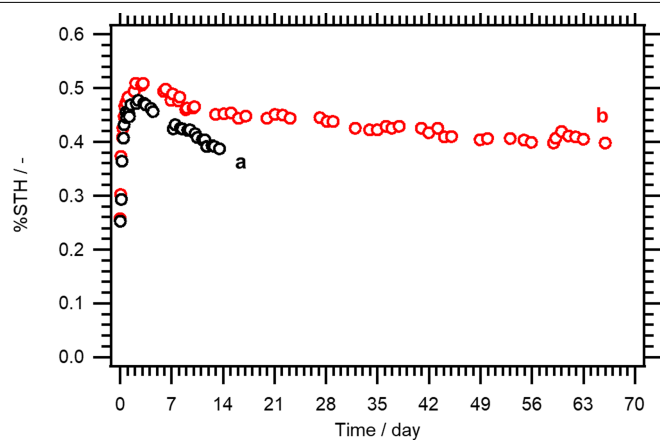
a, b, Cross-sectional backscattered electron images acquired by scanning electron microscopy of photocatalyst sheets prepared using a manual sprayer on a flat, clear glass sheet (**a**) or a program-controlled sprayer on a frosted glass

sheet (**b**). **c, d,** Cross-sectional images of a photocatalyst deposited by manual spraying on a flat, clear glass sheet, acquired by transmission electron microscopy before (**c**) and after (**d**) a field test lasting approximately six months.

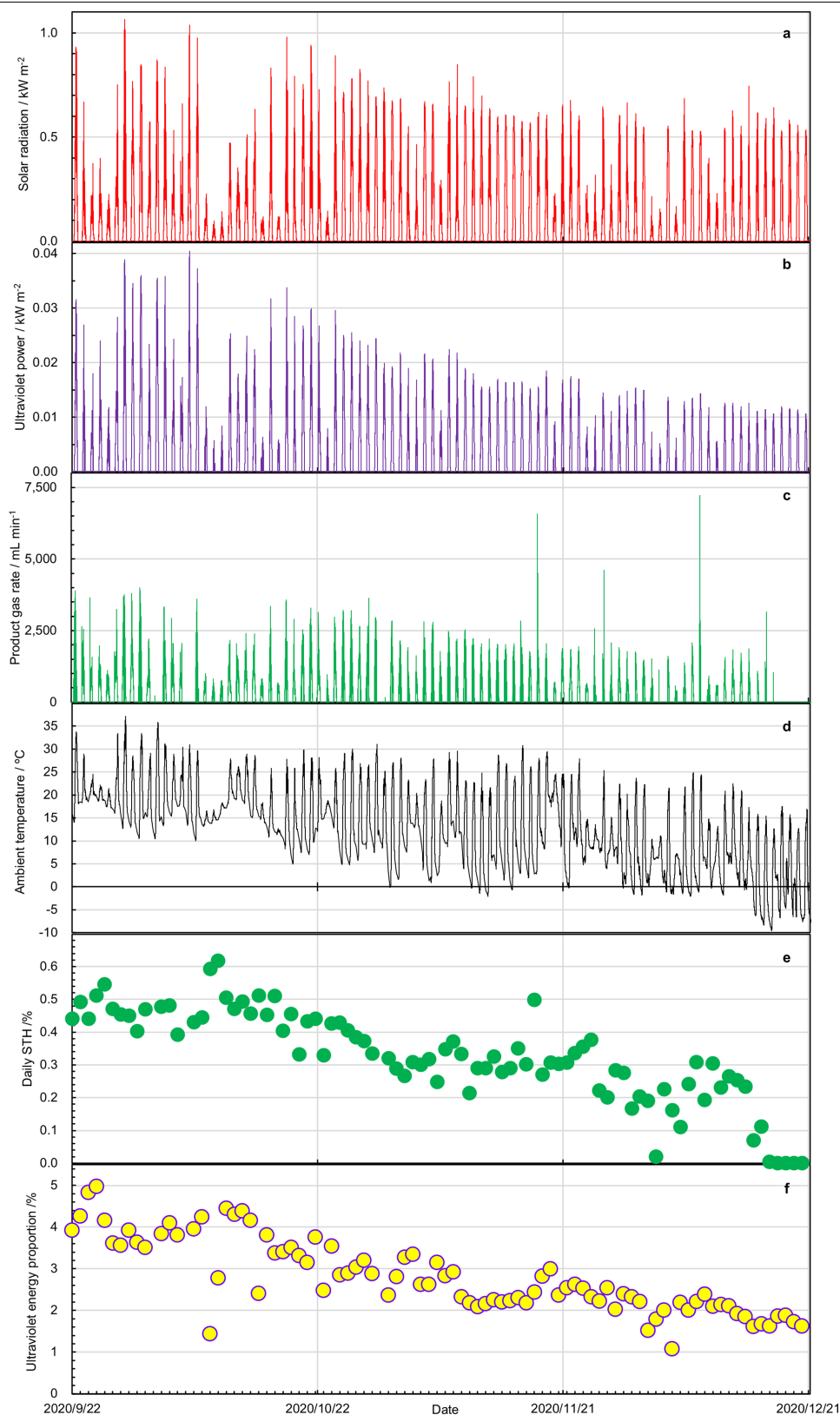


Extended Data Fig. 3 | The dependence of the water splitting rate of the photocatalyst sheet on the loading amount of the photocatalyst layer. Small photocatalyst sheets (5 cm × 5 cm) prepared on the flat, clear glass were

irradiated with the ultraviolet light emitting diode array. The loading amount of the photocatalyst layer is expressed as a relative value to the case where the modified SrTiO₃:Al loading is 0.89 mg cm⁻².

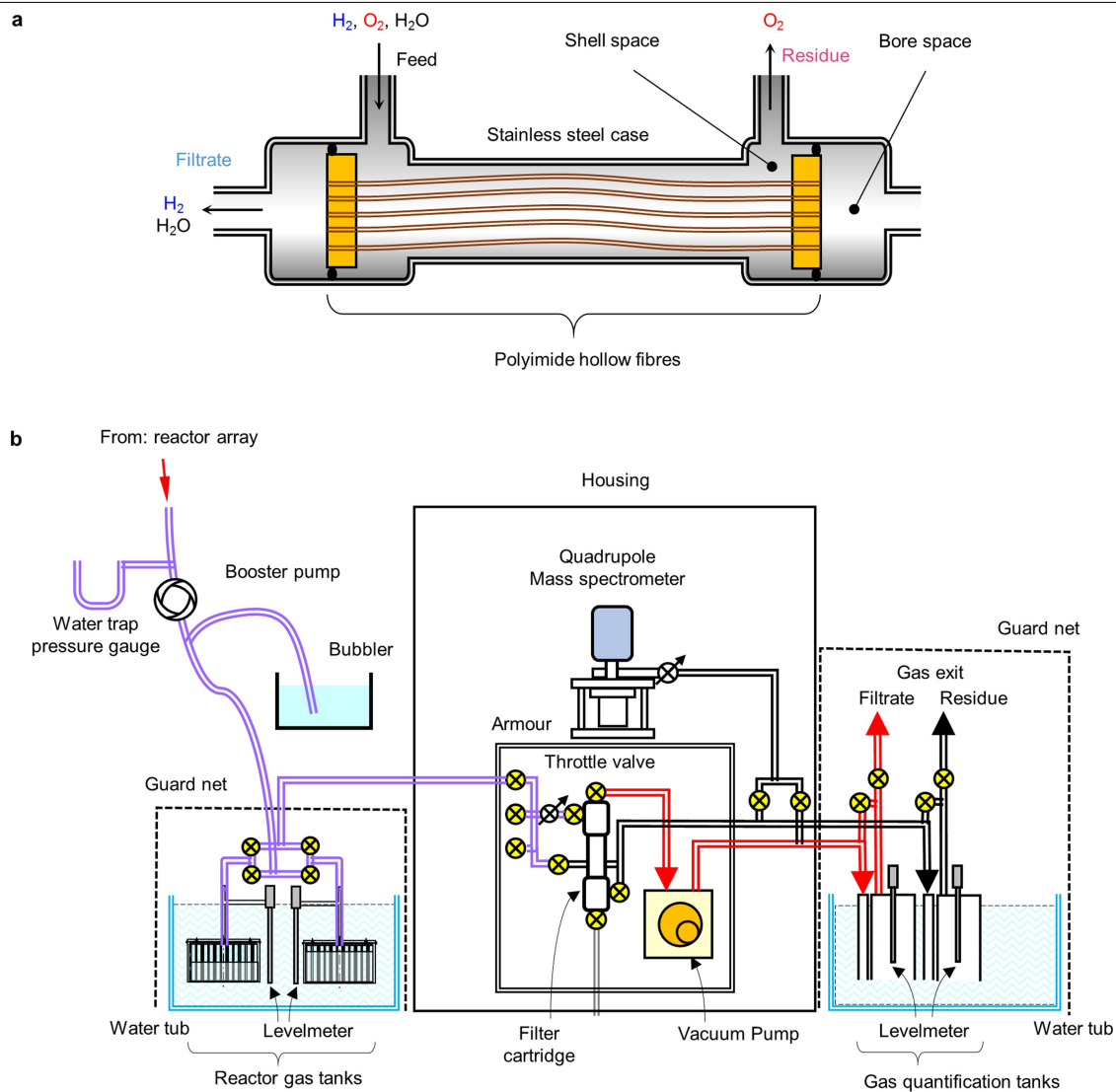


Extended Data Fig. 4 | Durability of photocatalyst sheets. a, b, The STH values of small photocatalyst sheets (25 cm^2) prepared using manual spraying with flat, clear glass (**a**) and a program-controlled sprayer with frosted glass (**b**) during the overall water splitting reaction under continuous irradiation with simulated standard sunlight.



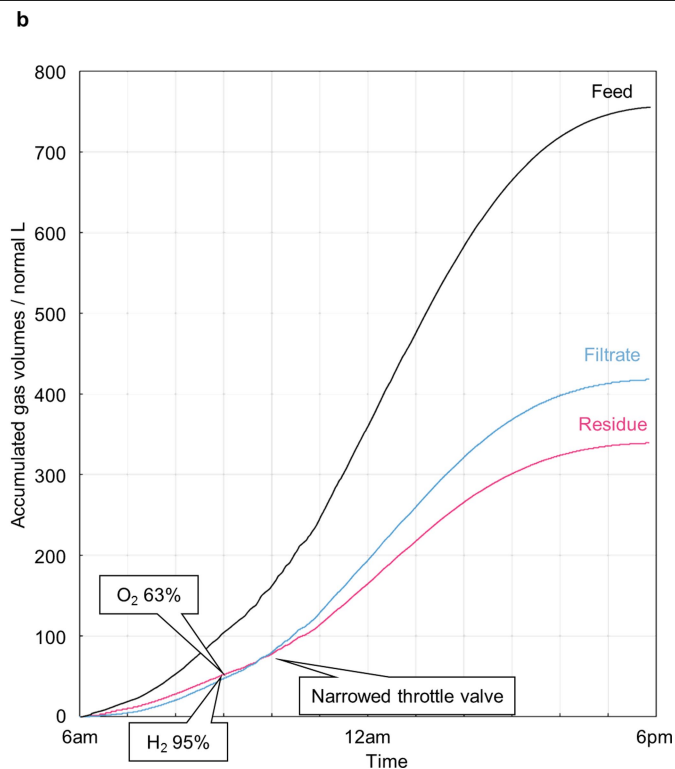
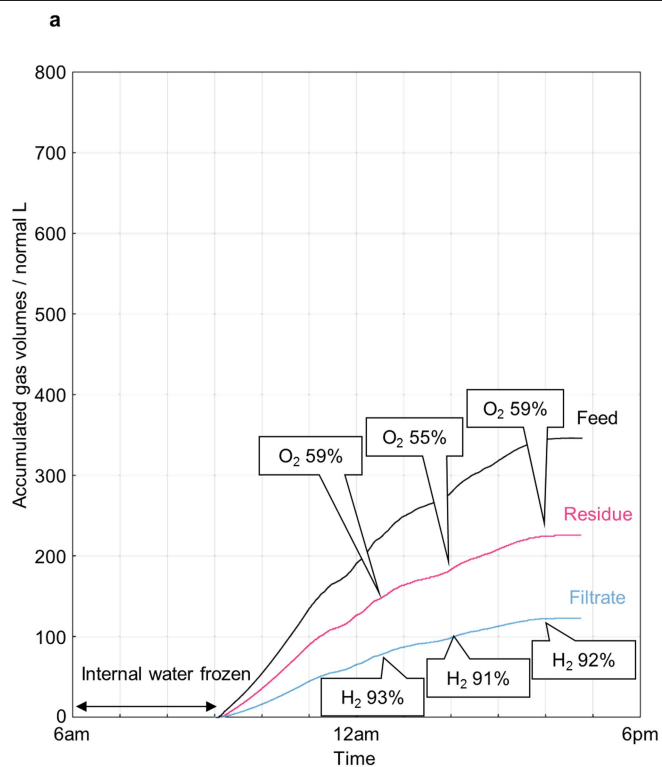
Extended Data Fig. 5 | Long-term records of the field test of the 100-m² photocatalyst panel reactor at Kakioka Research Facility from 22 September to 21 December 2020. a–f, Solar radiation (a), ultraviolet power (b), oxyhydrogen gas production rate (c), ambient temperature (d), daily STH value (e) and proportion of ultraviolet energy (f). The oxyhydrogen gas was not produced in late December 2020, because of the cold weather freezing the panel reactor units and other equipment. On 2, 6 and 30 October, the gas

was directed to the gas separation equipment, without delivery to the gas flowmeter. On 9 and 10 October, the daily STH values were abnormally high. These values are not reliable because of bad weather and the low solar radiation. On 9 and 10 October, the daily STH values were abnormally high. These values are not reliable because of bad weather and the low solar radiation. The data on 18 November and 8 December were not reliable either, owing to malfunction of the soap-film flowmeter.



Extended Data Fig. 6 | Structure of the gas separating facility. a. Schematic of the gas filtration cartridge. **b.** Schematic of the oxyhydrogen gas separation apparatus. The routes for the incoming oxyhydrogen gas, hydrogen-enriched

filtrate gas and oxygen-enriched residual gas are shown in purple, red and black, respectively.



Extended Data Fig. 7 | Performance of the gas separation unit connected to the 100-m² water splitting photocatalyst panel reactor. The hydrogen concentration in the filtrate gas and the oxygen concentration in the residue gas are intermittently notated along the curves. **a, b**, The data were acquired on

a day with mixed sun and cloud (6 February 2020; **a**) and a consistently sunny day (1 May 2020; **b**) at the Kakioka Research Facility. Note that the throttle valve was partially closed at 10:00 am on 1 May 2020 to reduce the gas feed rate.



Extended Data Fig. 8 | Ignition test of the panel reactor. The gas in the area indicated within the red square, with a light receiving area of 70 m^2 , was ignited by a spark. The spark gap was an assembly of two needle electrodes facing each

other within a tube. The distance between the electrodes was adjusted at approximately 0.5 mm . The ignition voltage was 15-kV a.c. and pulsed for approximately 1 s .